Deterministic Spiral Sparsity as a Geometric Prior for Sparse Neural Networks

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Transparency and Provenance Note

This document is an early research draft built around a simple personal idea: since the golden ratio is a fixed mathematical constant, it may be useful as a deterministic rule for placing sparse neural connections. The author does not claim prior expertise in sparse neural network theory and presents this work as an exploratory experiment rather than a settled scientific result.

Parts of the brainstorming, code scaffolding, analysis, and manuscript drafting were produced with the assistance of large language models, including ChatGPT, Gemini, and Grok, and were then selected, checked, and edited by the author. The main purpose of including this note is transparency: the underlying idea came from the author, while LLMs were used as exploratory and writing tools to help turn that idea into runnable experiments and a readable draft.

For additional transparency, the author provides the originating LLM conversations and notebook used for the experiments. The following are the final links:

- Summarized ChatGPT conversation: `https://chatgpt.com/share/69df5577-33b0-8322-ba64-ce369a5bbc66`
- Originating Gemini conversation: `https://gemini.google.com/share/74138edea74a`
- Originating Grok conversation: `https://grok.com/share/bGVnYWN5LWNvcHk_edc01472-898c-4d64-8693-ed93e653c62f`
- Experiment notebook / Colab: `https://colab.research.google.com/drive/146-IGZyJIjhASTcJM3qeD4QWtSd-EGqW?usp=sharing`

This manuscript should not be interpreted as a peer-reviewed claim of novelty, priority, or superior performance. Readers should independently verify experiments, compare against prior work, and treat the present results as an initial proof-of-concept study.

Abstract

This paper studies *deterministic spiral sparsity*, a structured sparsity prior generated by mapping points from a Fermat spiral onto a weight matrix before training begins. In the experiments, the golden angle is used as one concrete instantiation of this broader deterministic spiral construction. Unlike unstructured pruning, which removes parameters after optimization, or random sparse initialization, which relies on stochastic placement, the proposed approach uses a simple geometric rule to define which weights are trainable. We present the construction of the mask, describe masked linear and convolutional implementations, and report multi-seed studies on MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100. In the matched-budget MNIST comparison at 5% density, the recovered five-seed summary gives a final test accuracy of $94.216 \pm 0.166\%$ for the phyllotactic model, compared with $93.740 \pm 0.265\%$ for a random sparse baseline of matched density and $97.730 \pm 0.209\%$ for a dense model. A broader density sweep shows that the golden-angle mask is competitive overall and particularly strong on MNIST at 2% density, where it substantially outperforms the random sparse baseline, while Fashion-MNIST results are more mixed. On CIFAR-10 with a small CNN at 5% density, the golden-angle mask reaches $54.972 \pm 0.692\%$ test accuracy versus $54.856 \pm 0.922\%$ for matched random sparsity and $72.120 \pm 0.617\%$ for a dense baseline, indicating viability but no meaningful advantage over random sparsity on that harder setting. On CIFAR-100 with a Tiny ResNet at 10% density, the golden-angle mask reaches $27.187 \pm 2.120\%$, trailing both matched random sparsity at $29.017 \pm 1.240\%$ and a dense baseline at $37.450 \pm 1.205\%$. A follow-up training-budget stress test on the same CIFAR-100 setting sharpens that conclusion: golden and random sparsity are nearly tied after one epoch, but random sparse pulls ahead by epoch 3 and finishes 2.690 percentage points higher after 10 epochs, suggesting that the geometric mask is learning steadily but not catching up with training time alone. A revised CIFAR-10 edge microbenchmark with a tiny sparse CNN, a dense baseline, only 20 training steps, and five seeds across 1%, 2%, and 5% density shows that the phyllotactic mask is helpful only in the tightest regime: it beats random sparsity at 1% and 2% density, but loses at 5%. A companion dead-neuron diagnostic shows that the deterministic masks do not avoid disconnected units in any dramatic way. Follow-up causal and uniformity ablations on MNIST further show that neither maximizing coverage nor increasing entropy is sufficient to reproduce the small advantage of the original phyllotactic mask. Taken together, the evidence supports a narrower claim than the original golden-ratio hypothesis: the useful property is structured, non-repeating connectivity geometry rather than any specific constant such as the golden ratio or any simple scalar statistic. A final fair MNIST 2% rerun that adds a magnitude-pruned baseline further sharpens that interpretation: the golden original mask clearly outperforms matched random sparsity ($89.132 \pm 0.468\%$ versus $80.674 \pm 4.060\%$), but remains well below magnitude pruning ($96.540 \pm 0.131\%$) and the dense baseline ($97.564 \pm 0.127\%$). At this stage, the contribution is best understood as a promising architectural prior rather than a validated replacement for dense connectivity.

1 Introduction

Modern neural networks often achieve strong performance by using a large number of parameters, even when many of those parameters may be redundant. This observation motivates the study of sparse architectures, parameter pruning, and structured compression. Most existing approaches, however, either begin from dense parameterizations and remove weights later, or impose sparsity through random masks, thresholding, or hardware-driven structured blocks.

This work explores a different design principle: instead of viewing sparsity as a constraint imposed after the fact, we treat it as a *geometric prior* specified before training. The initial motivation comes from phyllotaxis, but the experiments in this paper are better interpreted as a study of deterministic spiral masks more generally. In practice, we sample points along a Fermat spiral and project them onto the coordinates of a weight matrix to define a fixed sparse mask.

The resulting architecture, which we refer to as *deterministic spiral sparsity* in its broad form, can be summarized by three features:

- the sparsity pattern is deterministic rather than random;
- the mask is fixed before training rather than learned by pruning;
- the pattern is generated by a compact geometric rule rather than stored as an arbitrary binary tensor.

The motivating question is straightforward: can a geometrically organized sparse network compete with more conventional sparse constructions, or exhibit desirable properties such as robustness, stability, or favorable scaling with very few trainable parameters? This paper does not claim a definitive answer yet. Instead, it formalizes the idea, provides an implementable model, and frames a research agenda for testing the hypothesis.

The main contribution of this draft is not a claim of universal superiority. Instead, it is an empirical identification of the regime in which deterministic geometric sparsity is most useful, together with a mechanism-level interpretation. The latest ablations show that simple explanations based on maximizing coverage or distribution entropy are insufficient: forced-coverage masks and more uniform random masks do not reproduce the original spiral advantage. The paper therefore studies deterministic spiral sparsity as a structured connectivity prior, not as a golden-ratio theorem.

2 Conceptual Motivation

In natural systems, phyllotactic arrangements are often associated with efficient spatial distribution. A common mathematical idealization uses the golden ratio

$$\varphi = \frac{1+\sqrt{5}}{2} \tag{1}$$

and the golden angle

$$\theta_g = 2\pi \left(1 - \frac{1}{\varphi}\right) \approx 137.5^\circ. \tag{2}$$

If points are placed sequentially with angular increment θ_g and radial growth proportional to $\sqrt{i}$, the resulting configuration avoids obvious radial alignments and tends to spread samples across a disc. We borrow this recipe as an organizing prior for neural connectivity. Informally, each potential weight location is treated as part of a two-dimensional field, and only matrix entries intersected by the spiral construction are activated.

The analogy should be interpreted carefully. The method does not prove that biological packing principles are optimal for learning. Rather, it offers an intuitively structured alternative to random sparse placement and a compact mathematical rule for generating reproducible masks.

3 Related Work

This paper sits at the intersection of sparse neural networks, structured connectivity design, and deterministic architectural priors. A large fraction of the sparsity literature begins from dense parameterizations and then removes or suppresses weights through pruning, thresholding, or learned saliency criteria. In that family, the central question is often which existing weights can be deleted with minimal damage after or during training. The present work asks a different question: if one must choose a sparse connectivity pattern *before* training, can a simple geometric rule provide a useful prior?

The most direct comparison in this draft is therefore not to post hoc compression, but to *matched-budget random sparsity*. That baseline isolates whether deterministic placement itself carries useful inductive bias when the number of active connections is held fixed. This framing is intentionally narrow. It does not imply that random sparsity is the strongest possible sparse baseline overall, nor that pruning-based methods, learned masks, Erdős–Rényi-style connectivity rules, or hardware-oriented structured sparsity are unimportant. Rather, random sparsity is used here as the cleanest first control for the geometric hypothesis.

The work also relates to broader ideas about structured wiring and deterministic initialization. Many architectures impose regularity through convolution, recurrence, low-rank structure, block sparsity, or hand-designed topologies motivated by efficiency or inductive bias. Deterministic spiral sparsity belongs to that broader family of *architecture-level priors*: instead of learning the wiring pattern, it fixes the pattern using a compact mathematical rule. What distinguishes the present proposal is not merely that it is sparse, but that it uses a non-periodic geometric construction intended to spread active connections broadly while avoiding obvious repeated overlap.

Finally, the paper should be read as complementary to, rather than competitive with, the strongest pruning and sparse-training literatures at this stage. Those lines of work typically optimize sparsity patterns more aggressively and at much larger scale. The contribution here is earlier-stage and more modest: to test whether a deterministic phyllotactic rule is viable at all, to identify where it appears most promising, and to clarify whether the golden-angle story survives contact with matched-budget experiments. That is why the main empirical emphasis falls on density dependence, dead-neuron diagnostics, and comparisons against random sparse baselines rather than claims of state-of-the-art compression.

A useful way to interpret the present work is as a study of sparse connectivity geometry rather than sparse optimization. The experiments are not intended to compete with the strongest pruning or learned-mask methods; instead, they isolate whether the spatial arrangement of fixed sparse connections matters under a controlled budget. That framing makes the present results complementary to pruning and sparse-training work, because it emphasizes coverage geometry, overlap, and connectivity regularity as explanatory variables.

4 Method

Consider a linear layer with input dimension d_{in} and output dimension d_{out}. Its dense weight matrix is

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}. \tag{3}$$

We define a binary mask

$$M \in \{0,1\}^{d_{\text{out}} \times d_{\text{in}}} \tag{4}$$

using a deterministic spiral sampling rule. For a target density $\rho \in (0,1]$, let

$$k = \lfloor \rho \, d_{\text{out}} d_{\text{in}} \rfloor \tag{5}$$

denote the target number of active weights. For each index $i \in \{0, 1, \dots, k-1\}$, define

$$r_i = \frac{\sqrt{i}}{\sqrt{k}}, \tag{6}$$

$$\theta_i = i \, \theta_g. \tag{7}$$

These polar coordinates are mapped into matrix coordinates by

$$x_i = \lfloor (0.5 + 0.5 r_i \cos\theta_i)\,(d_{\text{out}} - 1) \rfloor, \tag{8}$$

$$y_i = \lfloor (0.5 + 0.5 r_i \sin\theta_i)\,(d_{\text{in}} - 1) \rfloor. \tag{9}$$

The binary mask is then formed by setting

$$M_{x_i, y_i} = 1 \tag{10}$$

for all sampled indices, with all other entries equal to zero. In the exact-density implementation used for the main recovered experiments, collisions are resolved deterministically and any missing active positions are filled by a fixed fallback rule so that the final mask contains exactly k active entries.

The effective weight matrix is the Hadamard product

$$\widetilde{W} = W \odot M, \tag{11}$$

and the layer output for input vector h is

$$f(h) = \widetilde{W} h. \tag{12}$$

This construction produces a fixed sparse linear layer with trainable values only at locations selected by the geometric mask.

In this paper, the mask is treated as a fixed architectural prior, so the main question is not whether the mask can be learned, but whether its geometry produces a more useful sparse connectivity pattern under a fixed budget than a matched random mask.

4.1 Deterministic Spiral Mask

The initial implementation uses the following procedure:

1. create an all-zero mask for a given weight matrix shape;
2. compute the golden angle from the golden ratio;
3. generate a Fermat spiral over $k = \lfloor \rho\, d_{\text{out}} d_{\text{in}} \rfloor$ target active positions;
4. map the spiral coordinates to matrix indices;
5. multiply the dense parameter matrix by the fixed mask during the forward pass.

In the improved experiment, the sparse implementation is made more rigorous in two ways. First, the deterministic spiral mask is constructed to match the target density exactly rather than approximately, using collision handling and a deterministic fallback rule when two spiral samples land on the same matrix cell. Second, the random baseline is forced to use the exact same number of active connections, making the comparison a true matched-budget test rather than a comparison of approximately similar sparsity levels.

For illustration, a two-layer multilayer perceptron with dimensions $(128, 64, 10)$ has

$$128 \cdot 64 + 64 \cdot 10 = 8{,}832 \tag{13}$$

possible weights. Under the exact-density construction, the number of active weights is determined directly by the chosen density ρ. For example, at $\rho = 0.05$ the target active count is approximately $0.05 \times 8{,}832 \approx 442$ weights, before adding biases.

Even this simple example already illustrates the central point: the method imposes a substantial structural bottleneck rather than a mild regularizer. Any empirical success under this regime is therefore notable, while failure is still informative about the limits of geometry-only sparse design.

5 Experimental Results

We conducted a multi-seed experiment on MNIST using the implementation provided in the project context. Three multilayer perceptrons were trained for five epochs using Adam with learning rate 10^{-3} and batch size 128:

- a phyllotactic sparse network using the proposed mask at density 0.05;
- a random sparse network with matched density;
- a standard dense network with the same layer widths.

The experimental protocol was designed to be deliberately fair. All three models used the same multilayer perceptron topology, the same optimizer, the same training duration, and the same data preprocessing. The sparse comparison was especially strict: the phyllotactic and random models were matched for active connection count, and both used identical bias structure. The scientific question being tested is therefore narrow and well-defined: whether a fixed geometric sparsity prior is preferable to an equally sparse random prior under the same training budget.

The architectures used layer widths $(784, 256, 128, 10)$ and were evaluated over five random seeds $\{0, 1, 2, 3, 4\}$. In addition to standard test accuracy, we evaluated robustness under additive Gaussian input noise with standard deviation $\sigma = 0.20$. For the sparse models, the number of active parameters in the forward pass was 12,131, including biases, compared with 235,146 active parameters for the dense network. Thus, the sparse models used only about 5.16% of the active forward parameters of the dense baseline.

Ignoring biases, the active sparse weight count is approximately

$$0.05 \cdot 784 \cdot 256 + 0.05 \cdot 256 \cdot 128 + 0.05 \cdot 128 \cdot 10 \approx 11{,}737, \tag{14}$$

compared with 234,752 dense weights in the corresponding fully connected model. This clarifies that the phyllotactic and random models were matched not only in nominal density but also in exact active parameter count.

Table 1 summarizes the final held-out results at epoch 5. In the recovered five-seed summary, the phyllotactic model outperformed the random sparse baseline on both clean and noisy test accuracy. On clean test data, the gain was 0.476 percentage points (94.216% versus 93.740%). Under noisy evaluation, the gain was 0.474 percentage points (93.878% versus 93.404%). The dense model remained clearly stronger overall, but did so with roughly nineteen times more active forward parameters.

Table 1: Final MNIST performance at epoch 5 over five seeds. Values are mean ± standard deviation in percent.

Model	Test Accuracy	Noisy Test Accuracy	Active Parameters
Phyllotactic Sparse	94.216 ± 0.166	93.878 ± 0.196	12,131
Random Sparse	93.740 ± 0.265	93.404 ± 0.184	12,131
Dense	97.730 ± 0.209	97.656 ± 0.186	235,146

Figure 1 shows the mean test-accuracy trajectories over training. The phyllotactic curve lies above the random sparse curve at every epoch, indicating that the advantage is not limited to the final checkpoint. Averaged over seeds, the phyllotactic model improved from 90.330% to 94.216% test accuracy over five epochs, whereas the random sparse model improved from 89.990% to 93.740%.

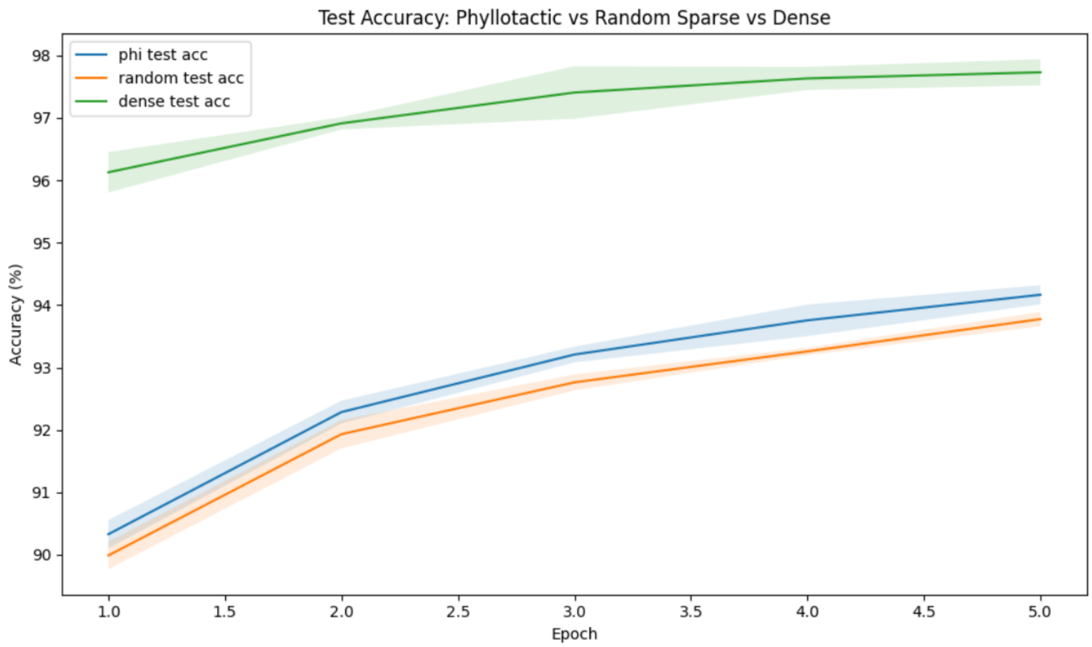

Figure 1: Mean MNIST test accuracy over five seeds for phyllotactic sparse, random sparse, and dense networks. Shaded bands indicate one standard deviation across seeds.

The central conclusion from this first multi-seed run is modest but encouraging: a fixed phyllotactic mask can slightly and consistently outperform a random sparse mask at matched density on both clean and noisy held-out evaluation, even though both sparse models remain behind a dense baseline. Put differently, structured wiring helps a little, randomness is slightly worse, and full density still wins clearly. This makes the idea more than a purely aesthetic construction, but not yet a demonstrated advance over conventional dense architectures.

A useful secondary perspective is relative efficiency. The phyllotactic model reaches about 96.36% of the dense model's final clean test accuracy while using only about 5.16% of the active forward parameters. That ratio does not make it superior to dense models in an absolute sense, but it does support the claim that geometry-informed sparse connectivity is a legitimate efficiency-oriented design direction.

5.1 Density Sweep Across Datasets

To test whether the 5% result generalizes across sparsity levels, we added a density sweep comparing golden-angle and random sparse masks at 1%, 2%, 5%, and 10% connectivity on MNIST, and at 2%, 5%, and 10% connectivity on Fashion-MNIST. Table 2 reports the recovered summary statistics. We omit Fashion-MNIST at 1% from the main table because only one valid run survived in the recovered summary, yielding an undefined cross-seed standard deviation and therefore not constituting a full multi-seed comparison.

Table 2: Recovered density-sweep results for golden-angle and random sparse masks. Values are mean ± standard deviation of final test accuracy in percent. Fashion-MNIST at 1% is omitted because the recovered summary contains only one valid run.

Dataset	Density	Golden-angle sparse	Random sparse
MNIST	1%	33.012 ± 6.465	32.960 ± 3.535
MNIST	2%	87.464 ± 0.334	74.864 ± 6.072
MNIST	5%	94.216 ± 0.166	93.740 ± 0.265
MNIST	10%	95.700 ± 0.085	95.486 ± 0.082
Fashion-MNIST	2%	75.162 ± 8.328	74.238 ± 6.909
Fashion-MNIST	5%	84.520 ± 0.068	84.634 ± 0.087
Fashion-MNIST	10%	85.546 ± 0.410	85.732 ± 0.261

The density sweep supports a more careful interpretation than the earliest draft. On MNIST, the golden-angle mask is modestly ahead at 1%, 5%, and 10%, and dramatically ahead at 2%, where it outperforms the matched random baseline by 12.600 percentage points. On Fashion-MNIST, the recovered results are more mixed: golden-angle sparsity is slightly better at 2%, but slightly worse at 5% and 10%. The cleanest synthesis is therefore not that golden-angle sparsity is uniformly superior, but that it is a competitive deterministic sparse prior overall, with one especially strong recovered advantage on MNIST at 2% density.

5.2 Performance at the Edge of Sparsity

The 2% MNIST condition is the most informative regime in the current paper because it sits near the edge where sparse connectivity becomes difficult to allocate well. At 10% density, both deterministic and random masks have enough active connections that the performance gap becomes small. At 2%, by contrast, the connection budget is tight enough that placement geometry matters more strongly. The present evidence suggests that this regime is where deterministic design appears most valuable, even though the exact causal mechanism is not captured by simple coverage or entropy statistics alone.

The 1% MNIST condition adds an additional cautionary signal. Both methods perform poorly there, and the recovered standard deviations are much larger than at higher densities, indicating that extreme sparsity produces unstable training and strong seed sensitivity. Rather than weakening the paper, this helps locate the practical boundary of the approach: below a certain connection budget, neither structured nor random sparsity yields reliably strong performance with the current architecture and training setup.

Two methodological cautions remain important. First, the present study is limited to MNIST and Fashion-MNIST with shallow multilayer perceptrons, so the findings may not transfer to harder datasets or convolutional and transformer architectures. Second, while several results now use five seeds, that still constitutes a relatively small sample for strong statistical claims. Third, the incomplete Fashion-MNIST 1% condition is a reminder that the recovered summaries are not equally mature across all settings.

We therefore treat these statistics as descriptive rather than definitive. The tables report means and standard deviations across seeds, and the paper now includes one small-sample paired significance analysis for the CIFAR-10 edge microbenchmark, but it still does not provide confidence intervals or broad formal testing across the main MNIST and Fashion-MNIST sweeps. Adding those analyses would strengthen the claim that the 2% MNIST advantage reflects a robust effect rather than a small-sample fluctuation.

5.3 MNIST 2% Final Fair Comparison with Magnitude Pruning

To make the strongest low-density MNIST result as fair and interpretable as possible, we ran one final controlled comparison entirely within a single script at 2% density over the same five seeds $\{0, 1, 2, 3, 4\}$. The compared conditions were a dense multilayer perceptron, a matched random sparse model, the original golden-angle sparse model, and a magnitude-pruned baseline. The dense, random sparse, and golden original models were each trained for five epochs under the same optimizer, batch size, data loaders, and seed handling. The magnitude-pruned baseline was intentionally stronger: for each seed it was first pretrained densely for three epochs, then globally pruned to the same 2% density, and finally fine-tuned for two additional epochs.

This experiment changes the interpretation of the 2% regime in an important way. Among *fixed-from-initialization* sparse models, the original golden mask remains clearly stronger than matched random sparsity. The five-seed means are $89.132 \pm 0.468\%$ for golden original versus $80.674 \pm 4.060\%$ for random sparse, a paired mean advantage of 8.458 percentage points with paired $t = 4.349$, $p = 0.012$, and Cohen's $d = 1.945$. However, once a stronger pruning-based baseline is introduced, magnitude pruning performs much better than either fixed sparse construction, reaching $96.540 \pm 0.131\%$, while the dense baseline reaches $97.564 \pm 0.127\%$.

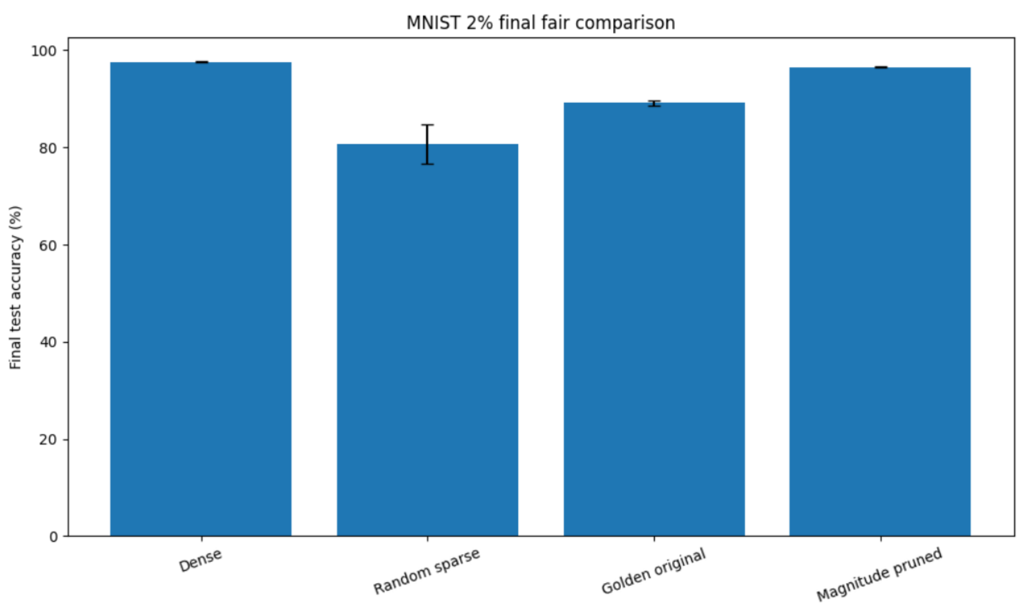

Figure 2: Final fair MNIST 2% comparison over five seeds. Golden original clearly outperforms matched random sparsity among fixed sparse masks, but both remain below magnitude pruning and the dense baseline. Error bars show one standard deviation across seeds.

The fair-comparison figure makes the hierarchy clear. Dense remains best, magnitude pruning is a very strong second, golden original is a solid third, and random sparse trails well behind. This is scientifically useful because it separates two different claims that earlier drafts could blur together. The first claim still survives: structured deterministic sparsity can be substantially better than matched random sparsity at a harsh 2% budget on MNIST. The second, stronger claim does not survive: fixed phyllotactic sparsity is not better than a strong pruning-based baseline when that baseline is allowed to discover a sparse mask after short dense pretraining.

That distinction improves the paper. It places the phyllotactic result in the right niche: not as the best overall way to obtain a 2% MNIST network, but as evidence that connectivity geometry matters *before training* and can produce a large gain over matched random placement even without post hoc mask optimization. Magnitude pruning remains the stronger practical baseline here, but it answers a somewhat different question because it benefits from information extracted during dense training before sparsification.

5.4 What the Results Do and Do Not Show

The current evidence supports a narrow claim: on MNIST and Fashion-MNIST, for these multilayer perceptrons and training setups, deterministic spiral sparsity can be competitive with and sometimes substantially better than matched random sparsity. That is a real empirical finding.

However, the combined experiments do not show that the golden ratio is universally special for intelligence, that irrational numbers as a class are strictly necessary, that golden-angle masks dominate every sparse baseline, or that geometry-based sparse networks can replace dense models in general. The new MNIST 2% fair comparison makes that especially clear: golden original beats matched random sparsity by a large margin, but loses decisively to a magnitude-pruned baseline and still remains below dense training. The contribution is better understood as evidence that connectivity pattern matters, that some deterministic angular rules are much better than others, and that geometric priors deserve to be studied alongside random and pruning-based sparsity methods. The aspect-ratio normalization ablation strengthens that interpretation by showing that visually correcting the rectangular distortion is not enough; the original spiral remains better because it preserves broader effective coverage.

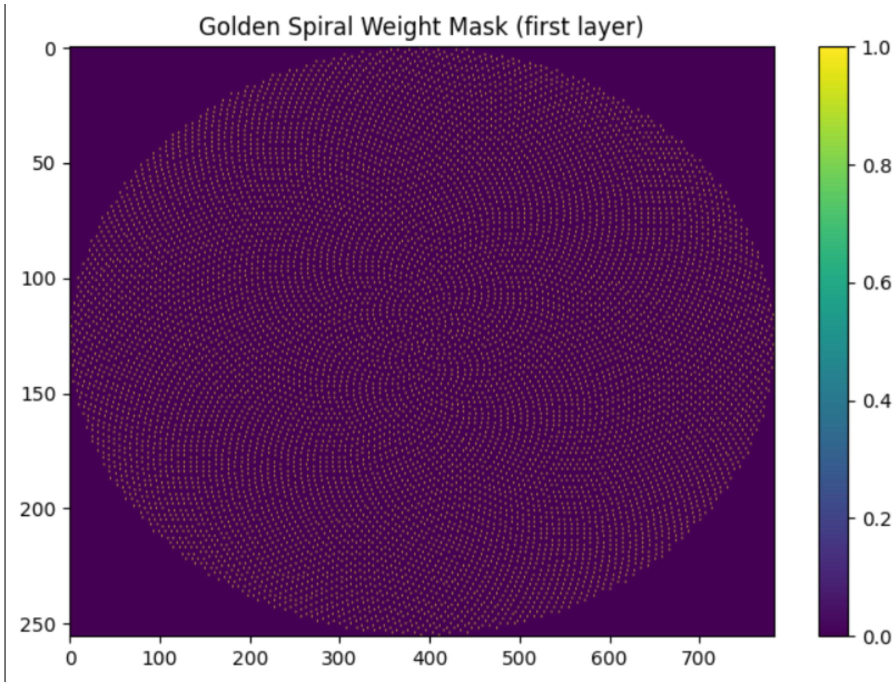

Figure 3: Visualization of the first-layer phyllotactic mask for the 784×256 layer under the improved exact-density construction. The pattern is highly structured and non-random, while still exhibiting aspect-ratio-induced anisotropy on the rectangular matrix.

Figure 3 confirms that the connectivity pattern is strongly organized rather than randomly dispersed. Interestingly, the image also reveals a limitation of the current implementation: when a circular spiral is projected onto a rectangular matrix, the pattern becomes anisotropic. The aspect-ratio-aware variant is informative because it shows that geometric normalization is not automatically beneficial. In the present results, the naive aspect-corrected mapping reduces input coverage and lowers accuracy relative to the original spiral on both MNIST and CIFAR-10. That suggests the relevant property is not simply making the spiral look more uniform in a rectangle, but preserving broad, effective coverage of active input channels. In other words, symmetry is not the goal; useful coverage is.

5.5 Angle Sensitivity Ablation

To probe whether the observed effect is specific to the golden angle or reflects a broader property of spiral-based deterministic sparsity, we ran a follow-up ablation over alternative angular increments. The compared models were: golden angle, e, $\sqrt{2}$, a seed-dependent random irrational angle, $\pi/2$ (listed as `deg90`), π (listed as both `deg180` and `pi`), a random sparse baseline, and a dense baseline. Each condition was trained for five epochs over five seeds on MNIST under the same 5% density setting.

Table 3 reports the final test accuracy. Three patterns stand out. First, the dense model remains the strongest overall. Second, several nontrivial spiral angles—golden, e, $\sqrt{2}$, and the sampled random-angle spiral—cluster closely together around 93.3–93.4% test accuracy, so no single angle in this group is clearly dominant in the present experiment. Third, simple rational angles such as π and $\pi/2$ perform dramatically worse, with π collapsing to near-chance performance and $\pi/2$ producing unstable and highly variable results.

Table 3: Angle ablation on MNIST over five seeds. Values are mean $\pm$ standard deviation of final test accuracy in percent.

Model / Angle	Test Accuracy
Dense	97.676 ± 0.186
Random Sparse	93.768 ± 0.176
e spiral	93.442 ± 0.097
Golden-angle spiral	93.440 ± 0.139
Random irrational-angle spiral	93.386 ± 0.284
$\sqrt{2}$ spiral	93.276 ± 0.211
$\pi/2$ spiral	36.254 ± 12.081
π spiral	11.350 ± 0.000

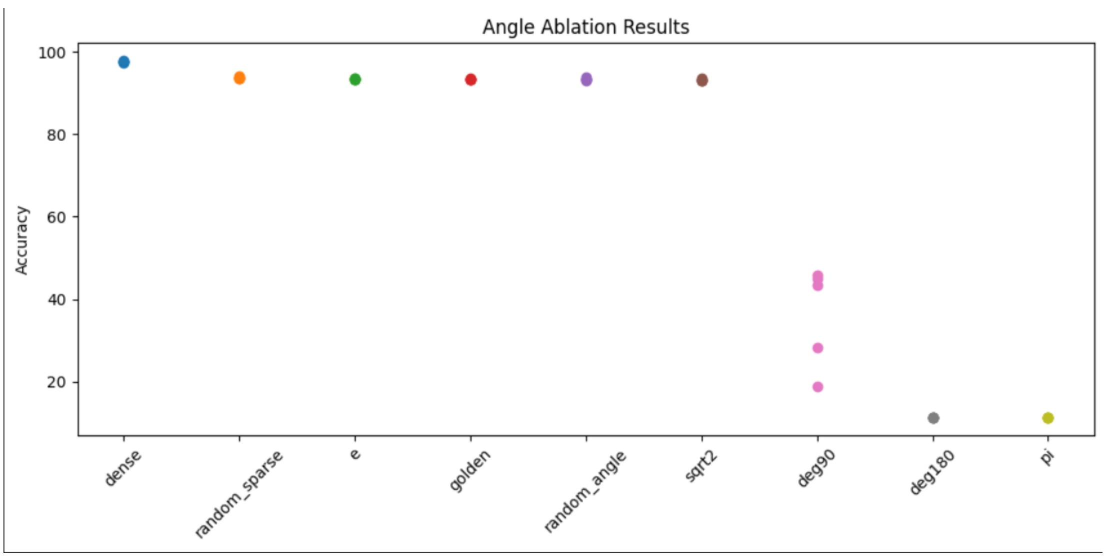

Figure 4: Final test accuracies across the angle ablation study. The results separate into a high-performing group of non-degenerate spiral angles, a weak but partially trainable $\pi/2$ group, and a fully degenerate π group near chance level.

This ablation materially sharpens the interpretation of the paper. The data no longer support the strongest version of a *golden-ratio-specific* claim. Instead, they support a broader and more defensible claim: deterministic spiral connectivity works best when the angular increment avoids short-period repetition, avoids repeated overlap, and distributes connections broadly across the layer. In that sense, the golden angle appears to be a strong member of a useful family of non-degenerate angular rules, rather than an isolated magical constant.

At the same time, the ablation reveals that geometry can act as a hard constraint. Bad angular choices produce catastrophic masks. The π and $\pi/2$ cases likely create spoke-like or repeatedly overlapping structures that undersample the input space and leave large regions effectively disconnected. This helps explain why some deterministic geometries are viable while others fail completely.

A final nuance is that, in this particular ablation code path, the random sparse baseline slightly outperformed the golden-angle model. That gap is small, and this ablation did not reuse exactly the same collision-handled mask-construction path as the earlier matched-budget study. This does not invalidate the earlier result; rather, it indicates that conclusions are sensitive to implementation details such as exact mask construction, collision handling, and the family of comparator masks. It also suggests that randomness may contribute a small stability benefit in addition to raw accuracy, since the random sparse baseline exhibits especially low variance in both MNIST and Fashion-MNIST. This sensitivity further motivates standardizing mask generation across all experiments. The safest synthesis across all experiments in this draft is therefore: structured deterministic geometry clearly matters, some angular rules are much better than others, and the golden angle is competitive but not uniquely dominant in the present evidence.

5.6 Generalization to Fashion-MNIST

To test whether the angle-ablation pattern transfers to a harder dataset, we repeated the same sparse-angle study on Fashion-MNIST using the same architecture, density, number of seeds, and training duration. The resulting ranking is shown in Table 4. The dense model again performs best overall, with mean test accuracy 87.892%. Among the sparse models, random sparsity is marginally strongest at 84.512%, followed extremely closely by sampled random-angle spiral (84.502%), golden angle (84.464%), e (84.356%), and $\sqrt{2}$ (84.262%).

Table 4: Fashion-MNIST angle ablation over five seeds. Values are mean $\pm$ standard deviation of final test accuracy in percent.

Model / Angle	Test Accuracy
Dense	87.892 ± 0.415
Random Sparse	84.512 ± 0.108
Sampled random-angle spiral	84.502 ± 0.263
Golden-angle spiral	84.464 ± 0.160
e spiral	84.356 ± 0.266
$\sqrt{2}$ spiral	84.262 ± 0.147
$\pi/2$ spiral	46.512 ± 7.868
π spiral	10.000 ± 0.000

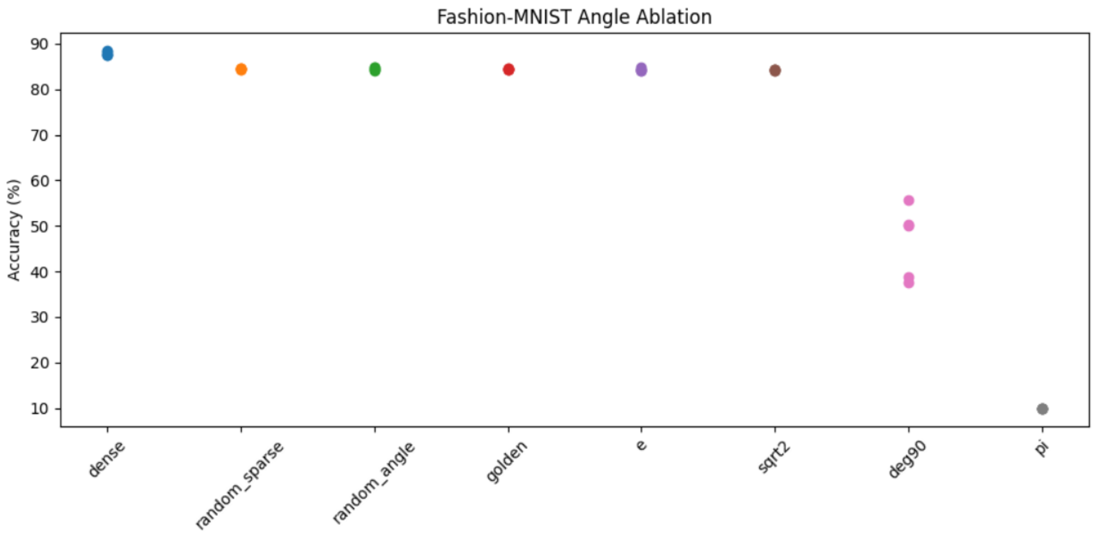

Figure 5: Final Fashion-MNIST test accuracies across the angle ablation study. As on MNIST, a high-performing cluster of non-degenerate spiral angles appears, while simple rational angles degrade sharply.

This second dataset materially strengthens the interpretation of the earlier MNIST ablation. The same qualitative structure reappears: a cluster of viable non-degenerate spiral angles, a weak and unstable $\pi/2$ condition, and a fully collapsed π condition at chance level. At the same time, the gap between golden angle and other good sparse baselines remains tiny. In particular, the golden-angle model is only 0.048 percentage points below random sparsity on average in this Fashion-MNIST run.

Taken together, the MNIST and Fashion-MNIST results support a more precise claim than the original hypothesis. The evidence does not indicate that the golden ratio is uniquely optimal, nor does it prove that irrationality by itself is the operative mechanism. Rather, it indicates that deterministic sparse connectivity benefits from angular rules that avoid short-period repetition and create broad layer coverage. The golden angle remains a natural and interpretable choice within that family, but the broader phenomenon appears to be about non-degenerate geometric spreading rather than golden-ratio exclusivity.

5.7 CIFAR-10 Small CNN

To test whether the method transfers beyond small multilayer perceptrons, we evaluated 5% sparse masks on CIFAR-10 using a small convolutional network with two convolutional layers, two linear layers, batch normalization, and dropout. The sparse models again used matched active-connection budgets, and all conditions were trained for five epochs over five seeds. Table 5 summarizes the final results.

Table 5: CIFAR-10 small-CNN results at 5% density over five seeds. Values are mean $\pm$ standard deviation of final test accuracy in percent.

Model	Test Accuracy	Train Accuracy
Phyllotactic Sparse	54.972 ± 0.692	46.073 ± 0.599
Random Sparse	54.856 ± 0.922	45.445 ± 0.759
Dense	72.120 ± 0.617	66.991 ± 0.350

The CIFAR-10 result is useful precisely because it is not a dramatic win. The phyllotactic and random sparse CNNs are essentially tied, with only a 0.116 percentage-point difference in mean test accuracy, while both remain far below the dense baseline. This indicates that deterministic geometric sparsity is viable on a harder vision task, but does not yet provide a meaningful advantage over matched random sparsity in this convolutional setting. In other words, the strong recovered MNIST result at 2% density does not automatically transfer to a more challenging dataset and architecture without further mask-design or architectural improvements.

Figure 6 shows the mean test-accuracy trajectories for the three CIFAR-10 models. The dense model remains clearly above both sparse curves throughout training, whereas the phyllotactic and random sparse models track one another closely. Figures 7 and 8 visualize the first convolutional-layer masks, highlighting the expected contrast between organized spiral structure and scattered random connectivity.

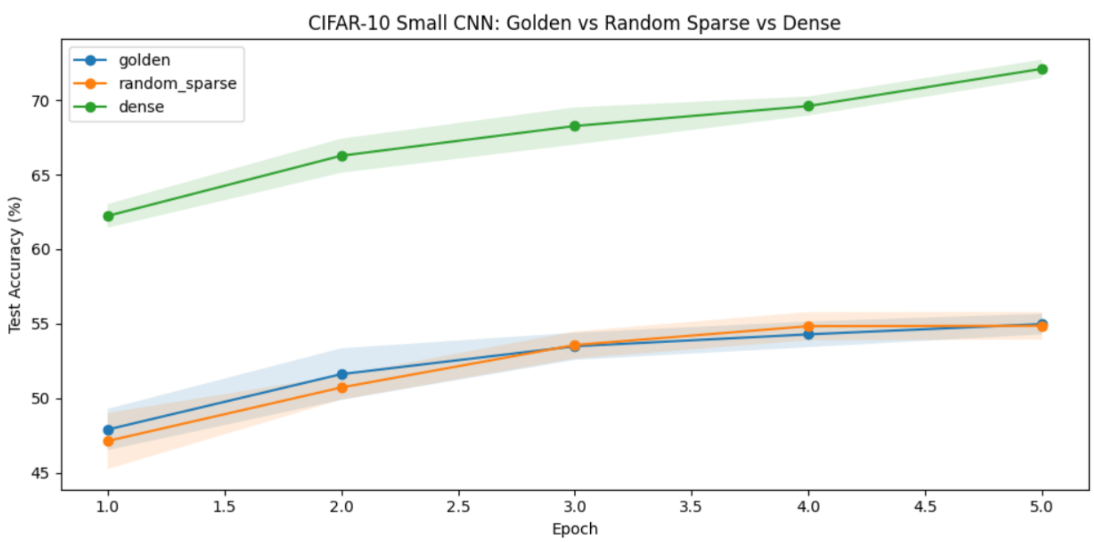

Figure 6: CIFAR-10 small-CNN mean test accuracy over five seeds for phyllotactic sparse, random sparse, and dense networks. The dense baseline learns substantially faster and reaches a much higher final accuracy, while the two sparse variants remain very close throughout training.

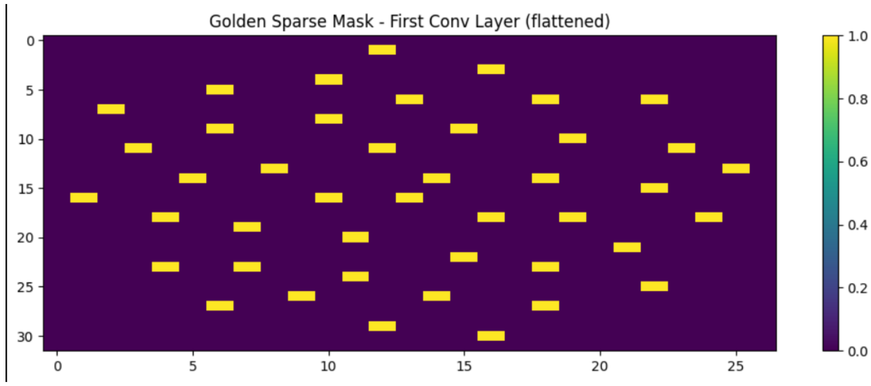

Figure 7: Flattened first-layer phyllotactic convolutional mask for the CIFAR-10 small CNN. The deterministic spiral structure remains visible even after flattening the convolutional receptive fields into matrix coordinates.

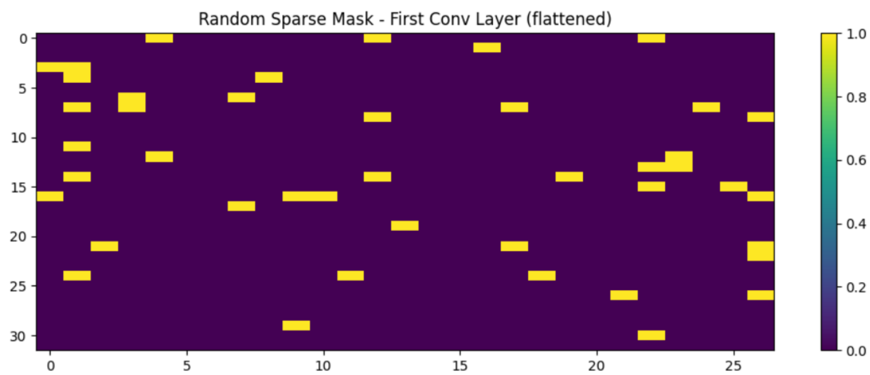

Figure 8: Flattened first-layer random sparse convolutional mask for the CIFAR-10 small CNN. In contrast to the phyllotactic mask, the active entries are scattered without obvious large-scale geometric organization.

5.8 CIFAR-100 Tiny ResNet

We next tested a harder vision setting by moving to CIFAR-100 and replacing the small CNN with a tiny residual network. In this experiment, the golden and random sparse models were both run at 10% density, again under matched active-connection budgets, for five epochs over three seeds. The goal was not to maximize absolute accuracy, but to determine whether the geometric prior remains competitive as task complexity and architectural depth increase. Table 6 reports the final results.

Table 6: CIFAR-100 Tiny-ResNet results at 10% density over three seeds. Values are mean ± standard deviation of final accuracy in percent.

Model	Test Accuracy	Train Accuracy
Phyllotactic Sparse	27.187 ± 2.120	28.124 ± 0.766
Random Sparse	29.017 ± 1.240	29.243 ± 0.249
Dense	37.450 ± 1.205	40.219 ± 0.209

Unlike the MNIST 2% result, the CIFAR-100 experiment does not favor the phyllotactic mask. Random sparsity outperforms the golden-angle construction by 1.830 percentage points in mean test accuracy, and the dense baseline remains substantially stronger than either sparse model. The gap is not enormous, but it is directionally consistent with the learning curves: on this harder 100-class task, the fixed spiral mask appears somewhat too rigid relative to a random sparse alternative. This strengthens the paper's central restraint. The geometric prior is not a universal winner; rather, it seems most promising in regimes where coverage is scarce but the task and architecture are still simple enough for deterministic structure to help rather than hinder learning.

Figure 9 shows the corresponding learning trajectories. The random sparse curve stays modestly above the phyllotactic curve throughout training, while the dense model separates clearly from both. Figures 10 and 11 show the stem-convolution masks used for the sparse Tiny-ResNet models.

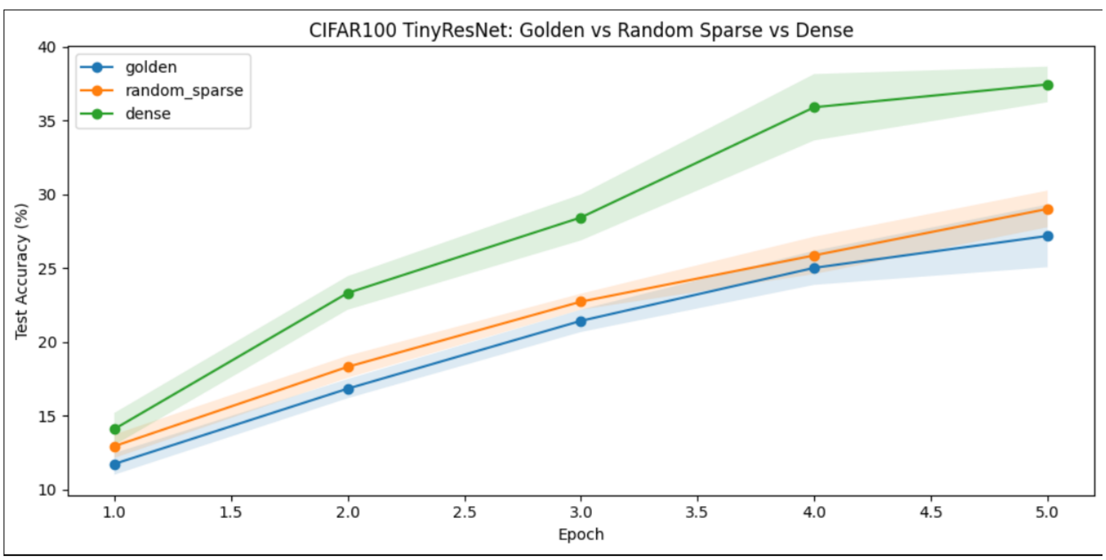

Figure 9: CIFAR-100 Tiny-ResNet mean test accuracy over three seeds for phyllotactic sparse, random sparse, and dense networks. Random sparsity stays slightly above the phyllotactic curve, while the dense baseline separates clearly from both.

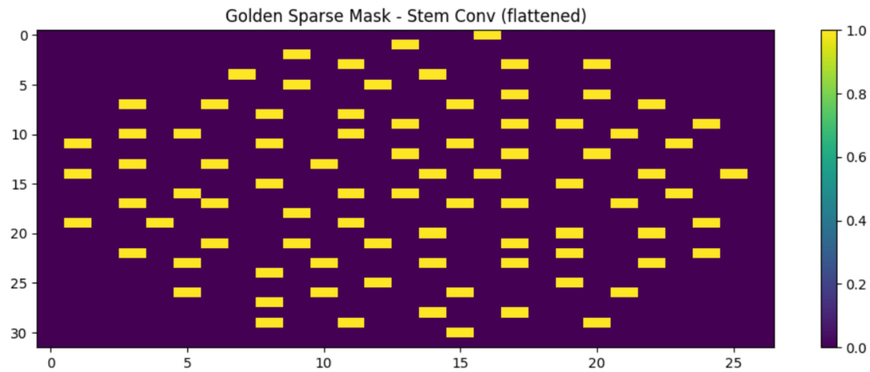

Figure 10: Flattened stem-convolution phyllotactic mask for the CIFAR-100 Tiny ResNet. The spiral organization is still clearly visible in the first layer.

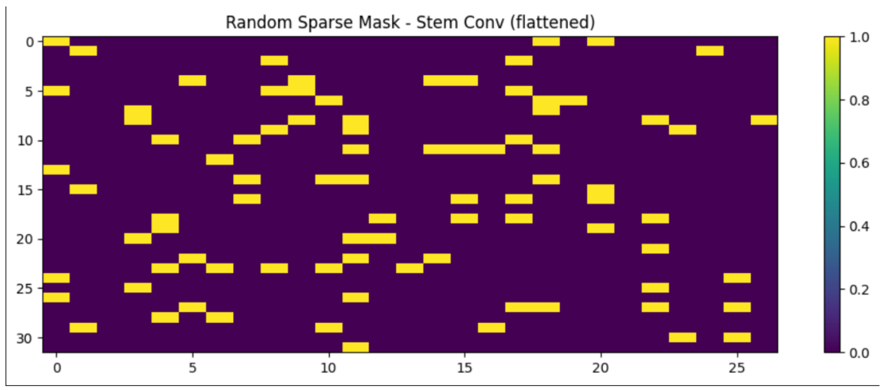

Figure 11: Flattened stem-convolution random sparse mask for the CIFAR-100 Tiny ResNet. The pattern lacks the large-scale structure seen in the phyllotactic version.

5.9 CIFAR-100 Training-Budget Stress Test

To distinguish between "golden is worse" and "golden is slower," we ran a follow-up budget-stress experiment on CIFAR-100 using the same 10% Tiny-ResNet setting but extending training to 10 epochs and recording checkpoints at epochs 1, 3, 5, and 10. This design keeps dataset, architecture, seed, initialization, and data order fixed within each run, so the main changing variable is training budget. Table 7 summarizes the checkpoint means across two seeds.

Table 7: CIFAR-100 Tiny-ResNet budget-stress results at 10% density over two seeds. Values are mean test accuracy in percent at matched checkpoints.

Model	Epoch 1	Epoch 3	Epoch 5	Epoch 10
Phyllotactic Sparse	11.890	18.395	24.470	32.235
Random Sparse	11.835	20.460	26.760	34.925
Dense	14.185	25.430	37.240	45.025

This stress test gives a clearer optimization picture than the earlier 5-epoch run alone. At epoch 1, the phyllotactic and random sparse models are essentially tied, with the geometric mask holding a trivial 0.055 percentage-point edge. By epoch 3, however, random sparsity has moved ahead by 2.065 points; by epoch 5 the gap is 2.290 points; and by epoch 10 it widens slightly further to 2.690 points. The key implication is that the phyllotactic mask is not merely slower and then convergent to the same solution. Instead, on this harder CIFAR-100 setting, extra training time does not erase the deficit. The geometric prior remains viable and improves steadily, but random sparsity appears to offer a modestly more favorable optimization path for this architecture and task.

Figure 12 shows the 10-epoch learning curves, and Figure 13 isolates the golden-versus-random checkpoint comparison. Together they show that the gap is small at initialization but becomes persistent once training begins in earnest. We also logged average gradient norms, but early automatic mixed-precision scaling occasionally produced `NaN` or `inf` summary values in the first few epochs, so we treat those diagnostics as exploratory rather than publication-grade evidence in the current draft.

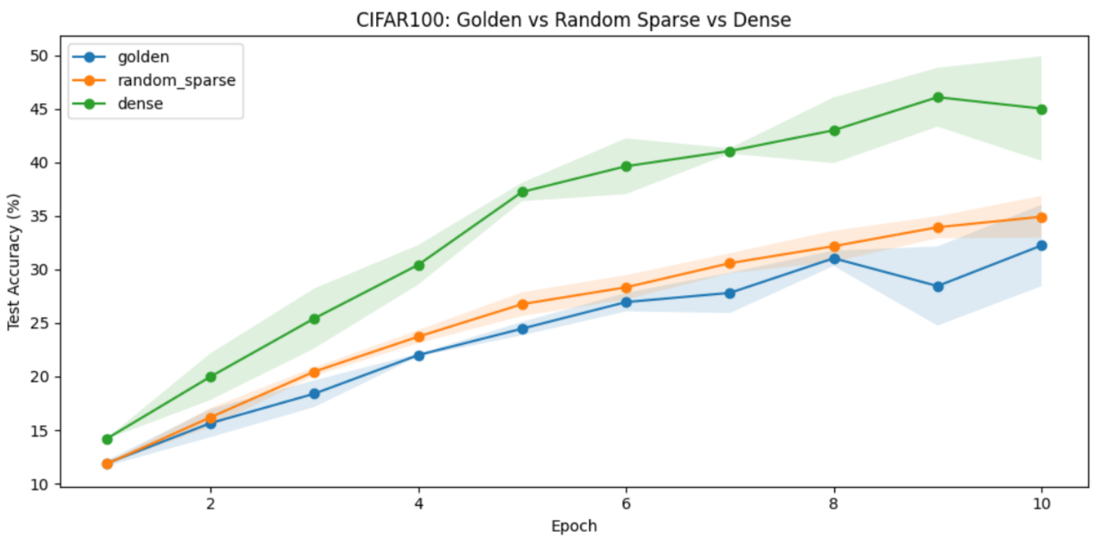

Figure 12: CIFAR-100 budget-stress learning curves over 10 epochs for phyllotactic sparse, random sparse, and dense Tiny-ResNet models. Random sparsity remains modestly above the phyllotactic curve throughout most of training, while the dense baseline separates clearly from both.

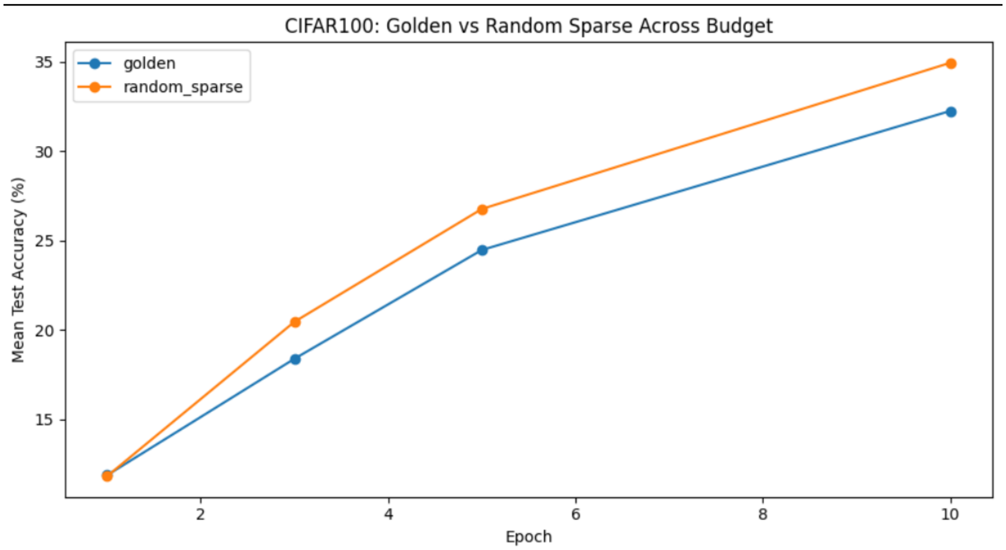

Figure 13: CIFAR-100 golden-versus-random checkpoint comparison under the training-budget stress test. The models are nearly tied at epoch 1, after which random sparsity develops a small but persistent advantage.

5.10 CIFAR-10 Edge Microbenchmark, Dense Baseline, and Dead-Neuron Diagnostic

As a deliberately harsh sanity check, we also ran a tiny CIFAR-10 edge-style microbenchmark designed to test whether a deterministic mask retains any advantage when both model capacity and training budget are extremely constrained. In this setting, we used the same TinyEdgeCNN architecture with three conditions: phyllotactic sparse, matched random sparse, and dense. All models were trained on a fixed small subset of CIFAR-10 for only 20 optimization steps, with checkpoints at steps 5, 10, and 20. We evaluated three densities, 1%, 2%, and 5%, and used five seeds at each density. Because the same subset, seed, and data order were used for each paired golden-versus-random run, the comparison isolates the effect of connectivity structure under a severe edge-like budget. Table 8 summarizes the final step-20 accuracy, Table 9 reports the paired golden-versus-random statistical tests, and Table 10 records the accompanying dead-neuron diagnostic.

Table 8: CIFAR-10 edge microbenchmark final step-20 results over five seeds per density. Values are mean ± standard deviation of test accuracy in percent, with the last column reporting the paired golden-minus-random difference in percentage points.

Density	Model	Step 20	Paired gap
1%	Dense	18.02 ± 0.87	—
1%	Phyllotactic Sparse	11.56 ± 1.59	$+1.76$
1%	Random Sparse	9.80 ± 0.14	—
2%	Dense	18.02 ± 0.87	—
2%	Phyllotactic Sparse	12.54 ± 1.96	$+0.66$
2%	Random Sparse	11.88 ± 1.56	—
5%	Dense	18.02 ± 0.87	—
5%	Phyllotactic Sparse	12.26 ± 1.68	-1.94
5%	Random Sparse	14.20 ± 0.97	—

These results should be interpreted cautiously. Even the dense model reaches only about 18% accuracy after 20 steps, so the microbenchmark is not evidence of practical competitiveness. However, it is still informative because it probes an extreme regime with vanishingly small connection budgets and almost no learning time. In that regime, the phyllotactic mask outperforms matched random sparsity at 1% density by 1.76 percentage points on average and at 2% density by 0.66 points, but loses by 1.94 points at 5%. This pattern fits the broader story of the paper more closely than the earlier two-density run: geometric determinism appears most promising at the tightest budgets, while random sparsity catches up or pulls ahead once the network is given a less severe connection budget. Dense remains clearly stronger than both sparse variants at every density.

The paired significance tests reinforce the need for restraint. With only five seeds, none of the golden-versus-random gaps are conventionally significant under a paired t-test: the p-values are 0.080 at 1%, 0.693 at 2%, and 0.155 at 5%. The corresponding Cohen's d values, 1.167, 0.212, and -0.874, suggest a potentially meaningful positive effect at 1%, a very weak effect at 2%, and a moderate negative effect at 5%, but the sample is too small to treat those as settled conclusions. The most defensible reading is therefore directional rather than confirmatory.

Table 9: Paired golden-versus-random statistical tests for the CIFAR-10 edge microbenchmark at step 20. The reported effect size is Cohen's d computed from the paired differences.

Density	Mean gap	Std. gap	t-statistic	p-value	Cohen's d
1%	$+1.76$	1.51	2.334	0.080	1.167
2%	$+0.66$	3.11	0.424	0.693	0.212
5%	-1.94	2.22	-1.747	0.155	-0.874

The dead-neuron diagnostic helps clarify one possible mechanism. Because the golden masks are deterministic, their dead-output fractions are fixed across seeds for a given density: 23.770% at 1% density, 11.475% at 2%, and 4.918% at 5%. The random masks produce very similar averages, $22.951 \pm 2.551\%$, $10.984 \pm 2.032\%$, and $2.951 \pm 2.032\%$, but with seed-dependent variability. In other words, the observed accuracy differences do not seem to arise from the golden mask simply avoiding dead units altogether. Instead, the more plausible interpretation is that both constructions face similar coarse connectivity bottlenecks at these densities, while the exact arrangement of the surviving paths changes optimization enough to produce small density-dependent swings.

Figure 14 summarizes the final step-20 hierarchy. Dense stays well above both sparse conditions throughout, golden has a modest edge over random at 1% and 2%, and random sparse takes over at 5%. The checkpoint summaries follow the same general pattern: at 1% density the phyllotactic model stays above random sparsity at steps 5, 10, and 20; at 2% density the ordering is mixed early but ends slightly pro-golden by step 20; and at 5% density the random mask is already ahead by step 5 and remains ahead thereafter. We therefore keep the current figure set focused on the updated final-accuracy summary and use Table 10 for the dead-neuron comparison.

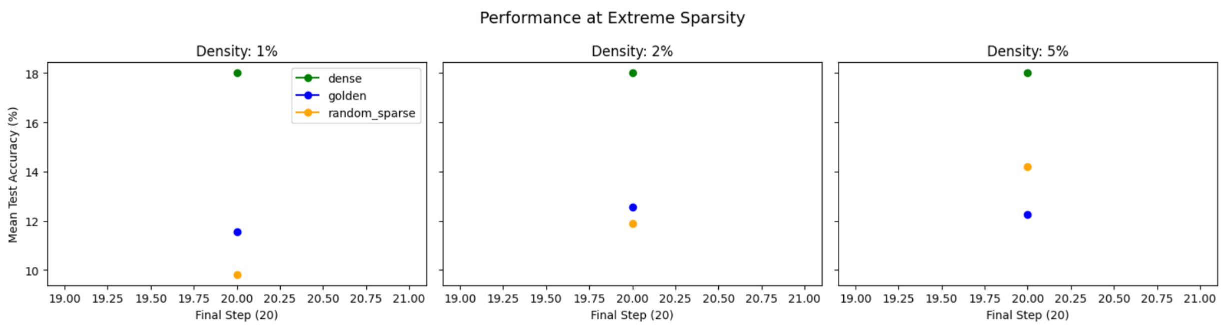

Figure 14: Compact summary of final step-20 accuracy across the three edge-microbenchmark densities. Dense remains clearly superior at all densities, while phyllotactic sparsity outperforms matched random sparsity at 1% and 2% but not at 5%.

Table 10: Dead-neuron diagnostic for the CIFAR-10 edge microbenchmark at step 20. Values are mean $\pm$ standard deviation of the fraction of output channels or neurons with zero active incoming masked weights.

Density and model	Dead neurons (%)	Overall sparsity (%)
1% golden	23.770 ± 0.000	99.005
1% random sparse	22.951 ± 2.551	99.005
2% golden	11.475 ± 0.000	97.997
2% random sparse	10.984 ± 2.032	97.997
5% golden	4.918 ± 0.000	94.998
5% random sparse	2.951 ± 2.032	94.998

6 Mechanism Analysis

6.1 Revisiting Coverage and Distribution Hypotheses

Initial analysis suggested that input coverage and distribution entropy correlate positively with performance. However, a controlled ablation reveals that these factors are not sufficient to explain the observed advantage of phyllotactic sparsity.

A coverage-forced variant, which guarantees 100% input coverage and eliminates dead outputs, does not outperform the original phyllotactic mask. This indicates that coverage alone is not the causal mechanism.

Similarly, direct measurement of row and column entropy and coefficient of variation shows that random sparse masks exhibit higher entropy and more uniform distributions than the phyllotactic mask, yet perform slightly worse in accuracy.

These results collectively falsify a simple explanation based on coverage or uniformity alone.

6.2 Evidence for Structured Non-Random Connectivity

Given that neither coverage nor uniformity explains the performance gap, the results point toward a different mechanism: structured, deterministic connectivity.

The phyllotactic mask imposes a low-discrepancy, spatially correlated pattern of connections. Unlike random sparsity, which produces independent and potentially clustered connections, the spiral construction introduces a form of structured irregularity: it avoids both extreme clustering and strict uniformity.

This suggests that the benefit of phyllotactic sparsity arises from its deterministic spatial correlations, which may improve gradient flow or feature mixing in ways that are not captured by simple statistical measures such as entropy or coverage.

Importantly, this interpretation is consistent with the observation that enforcing uniform coverage or maximizing entropy does not improve performance, indicating that the useful property lies in the geometry of connectivity rather than its marginal statistics.

6.3 What the Aspect-Aware Variant Shows

To test whether the visible anisotropy in rectangular mask projections could be corrected without sacrificing performance, we ran an aspect-ratio normalization ablation on the strongest low-density regime and on the small convolutional regime. The compared models were the original golden spiral, an aspect-aware golden spiral, matched random sparsity, and a dense baseline. We evaluated MNIST at 2% density and CIFAR-10 at 5% density over five seeds each, and we recorded input-coverage and dead-output diagnostics from the reconstructed masks.

Table 11 and Figure 15 summarize the outcome. The aspect-aware variant does not improve performance in the recovered runs. Instead, it lowers coverage and reduces accuracy. On MNIST, clean accuracy drops from 87.62% for the original golden spiral to 69.43% for the aspect-aware version, while random sparsity reaches 74.69%. On CIFAR-10, clean accuracy drops from 38.39% to 34.35%, with random sparsity at 38.71%. That result is important because it shows the paper is not simply about making the mask look more geometrically elegant. The useful property is effective connectivity coverage, not visual regularity.

Table 11: Aspect-ratio ablation results over five seeds. Accuracies are reported as mean final test accuracy in percent; the last two columns summarize the associated connectivity diagnostics.

Dataset	Model	Clean Acc. (%)	Noisy Acc. (%)	Input coverage (%)	Dead outputs (%)
MNIST (2%)	Golden original	87.62	86.64	69.72	0.91
MNIST (2%)	Golden aspect-aware	69.43	68.35	50.67	6.67
MNIST (2%)	Random sparse	74.69	73.26	70.44	2.26
MNIST (2%)	Dense	97.53	97.49	100.00	0.00
CIFAR-10 (5%)	Golden original	38.39	38.26	79.76	4.69
CIFAR-10 (5%)	Golden aspect-aware	34.35	34.31	59.03	7.03
CIFAR-10 (5%)	Random sparse	38.71	38.75	80.42	6.56
CIFAR-10 (5%)	Dense	57.98	57.05	100.00	0.00

The aspect-aware ablation remains informative, but not because it confirms a coverage-based mechanism. On MNIST, mean input coverage fell from 69.72% for the original spiral to 50.67% for the aspect-aware version, while dead outputs increased from 0.91% to 6.67%. On CIFAR-10, coverage fell from 79.76% to 59.03% and dead outputs rose from 4.69% to 7.03%. The key point is that geometric normalization is not automatically beneficial: making the mask look more regular in a rectangle can still degrade the structured spatial correlations that appear to matter for learning.

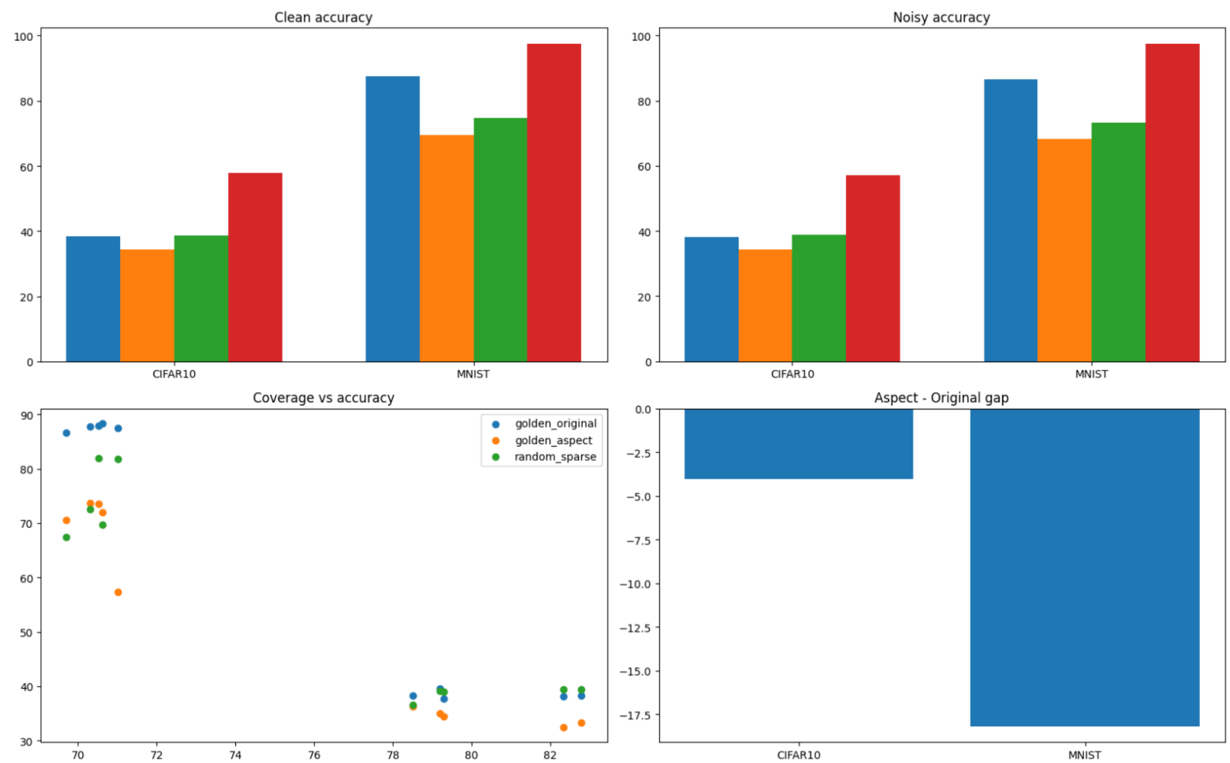

Figure 15: Mechanism analysis of phyllotactic sparsity. The aspect-aware variant underperforms the original spiral, and follow-up ablations show that neither maximizing coverage nor increasing distribution uniformity is sufficient to reproduce the phyllotactic advantage. The remaining evidence points toward structured connectivity geometry rather than simple scalar mask statistics.

6.4 What Can and Cannot Be Claimed

The present evidence supports a narrow claim: deterministic sparse geometry can help under extreme sparsity, but the effect is not explained by coverage or uniformity alone. The evidence does not support a universal claim that the golden ratio is optimal, or that any aspect-aware normalization is necessarily better. The effect appears regime-dependent and architecture-dependent.

6.5 Cross-Dataset Summary

Table 12 collects the main empirical message of the paper in one place. The most important pattern is not that the deterministic spiral mask always wins, but that its behavior depends strongly on regime. It is mildly favorable on the recovered MNIST matched-budget test, especially strong on the recovered MNIST 2% density condition, roughly tied on CIFAR-10 with a small CNN at 5% density, unfavorable on CIFAR-100, negative for the aspect-ratio-normalized variant, and mixed in the edge microbenchmark where it helps at 1% and 2% density but loses at 5%. That density dependence, together with the new mechanism ablations that rule out simple coverage and uniformity explanations, is the cleanest high-level takeaway from the current draft.

Table 12: Cross-dataset summary of the main matched-budget comparisons in the paper. Accuracies are final test accuracies in percent. The takeaway column states the intended interpretation rather than a claim of statistical significance.

Dataset	Family	Density	Spiral	Random	Dense	Takeaway
MNIST	MLP	5%	94.216	93.740	97.730	Original golden helps; aspect-aware hurts in separate ablation
MNIST	MLP	2%	87.464	74.864	—	Strongest regime for original golden
Fashion-MNIST	MLP	2–10%	mixed	mixed	—	Mixed overall; no consistent structured advantage
CIFAR-10	small CNN	5%	54.972	54.856	72.120	Random and original are close; aspect-aware is worse
CIFAR-100	Tiny ResNet	10%	27.187	29.017	37.450	Random wins; dense wins clearly
CIFAR-10 edge	tiny CNN	1%, 2%, 5%	11.56 / 12.54 / 12.26	9.80 / 11.88 / 14.20	18.02	Original helps only at the tightest budgets

7 Experimental Program

The next stage of this project should evaluate the method under controlled comparisons. A minimal experimental matrix is as follows.

7.1 Baselines

Each deterministic-spiral model should be compared against:

- a dense multilayer perceptron with the same layer widths;
- a random sparse network with the same number of active weights;
- a magnitude-pruned model compressed to the same final sparsity;
- optionally, a block-sparse or low-rank baseline if hardware efficiency becomes relevant.

7.2 Datasets

MNIST now serves as the first proof-of-concept dataset for the method, and the current revision includes recovered density-sweep comparisons across multiple sparsity levels, five-seed angle-ablation studies on MNIST and Fashion-MNIST, a five-seed fair MNIST 2% rerun that adds a magnitude-pruned baseline, a five-seed CIFAR-10 small-CNN transfer experiment, a three-seed CIFAR-100 Tiny-ResNet study, a two-seed CIFAR-100 training-budget stress test, and a five-seed CIFAR-10 edge microbenchmark with dense, random-sparse, and phyllotactic models at 1%, 2%, and 5% density, together with a dead-neuron diagnostic. Additional lightweight tabular or synthetic benchmarks remain useful because they permit repeated runs, ablation studies, and uncertainty estimates with low computational cost. The initial objective is not state-of-the-art performance, but a clean comparison between geometric sparsity and standard alternatives.

7.3 Metrics

Recommended evaluation metrics include:

- test accuracy and training loss;
- parameter count and effective sparsity ratio;
- convergence speed in steps or epochs;
- variance across random seeds;
- robustness under input noise, weight perturbation, or adversarial corruption.

The current manuscript now includes held-out test accuracy, repeated-seed variability, and one small-sample paired significance analysis for the edge microbenchmark, but it still lacks confusion matrices, confidence intervals across the broader experiment set, wall-clock time, and calibration measures.

7.4 Ablations

Several ablations naturally follow from the construction:

- replace the golden angle with other irrational or rational angular increments;
- vary the radial law from $\sqrt{i}$ to linear or logarithmic growth;
- alter the number of seeds independently of $\min(d_{\text{out}}, d_{\text{in}})$;
- compare fixed phyllotactic masks with trainable or partially trainable masks;
- test Fibonacci-inspired width schedules such as $(55, 34, 21)$ or deeper sequences.

These studies would help determine whether any observed benefit arises specifically from phyllotaxis, from determinism alone, or simply from the regularizing effect of extreme sparsity.

8 Discussion

The proposal can be interpreted in two complementary ways.

First, it is a compression prior. Because the mask is generated from a short rule, the architecture can in principle be reconstructed without storing an arbitrary sparsity pattern. This may be attractive in settings where reproducibility and compact architectural descriptions matter.

Second, it is a representational hypothesis. The mask imposes a non-random spatial ordering on which interactions between input and output units are allowed. If learning benefits from structured diversity rather than redundant connectivity, then a geometric prior could outperform a random sparse pattern at the same parameter budget.

These results position deterministic spiral sparsity as an intermediate approach between random initialization and learned sparsity methods. While it does not match the performance of magnitude pruning, it provides a zero-additional-compute, deterministic alternative that significantly improves over matched random sparsity in extreme low-density regimes. That makes it useful as a clean architectural prior even when it is not the strongest overall route to final accuracy.

There are also clear limitations. The current mapping from spiral coordinates to matrix indices is heuristic and sensitive to layer aspect ratio, yet the new aspect-ratio normalization ablation shows that this issue is subtler than it first appeared: visible anisotropy alone is not the main problem, because a direct normalization step degraded performance on both MNIST and CIFAR-10. Future geometric corrections will therefore need to preserve the useful structured correlations of the original mask rather than merely making it look more symmetric. The CIFAR-10 and CIFAR-100 experiments add a second limitation: even when deterministic sparse masks remain viable, their advantage over random sparsity may shrink or reverse on harder datasets and convolutional architectures. The CIFAR-100 budget-stress result makes this sharper still: the geometric mask does not simply lag early and then catch up with more epochs; rather, the random sparse advantage persists as training budget increases. The revised CIFAR-10 edge microbenchmark adds a complementary nuance: under an intentionally harsh low-budget regime, the original phyllotactic mask holds a small average edge at 1% and 2% density, but the effect reverses by 5%. The dead-neuron diagnostic suggests that this is not simply because the original mask avoids disconnected units, and the new causal and entropy ablations go further by showing that neither maximizing coverage nor enforcing uniformity reproduces the phyllotactic advantage. The more plausible explanation is therefore that the original spiral provides a structured non-random connectivity pattern whose useful properties are not captured by simple scalar statistics alone. Moreover, ordinary weight matrices do not possess an intrinsic Euclidean geometry in the same way that points in physical space do. The usefulness of the induced structure must therefore be demonstrated empirically rather than assumed from the biological metaphor. Finally, the sparse models still contain dense trainable tensors under the hood, so true systems-level memory and compute savings would require a more specialized implementation. The present evidence therefore speaks most directly to inductive bias and matched-budget learning behavior, not yet to end-to-end deployment efficiency. The method also appears to impose structural constraints that limit performance on more complex datasets such as CIFAR-100.

9 Conclusion

The revised evidence does not support a simple explanation based on input coverage or distribution entropy. Instead, the results suggest that the advantage of deterministic spiral sparsity arises from structured, deterministic connectivity patterns that differ fundamentally from random sparsity.

While the precise mechanism remains open, the experiments show that neither maximizing coverage nor enforcing uniformity is sufficient to reproduce the observed gains. This shifts the interpretation of deterministic spiral sparsity away from scalar statistical properties and toward structured connectivity geometry as the key factor.

The main contribution of this work is therefore not a validation of a specific geometric constant, but an empirical demonstration that the spatial organization of sparse connections can matter independently of coverage and distribution statistics.

The next useful step is not more training by default, but stronger ablation on connectivity geometry itself. The present results already show that simple coverage and entropy explanations are insufficient. Future work should ask whether structured low-discrepancy or blue-noise-style masks can reproduce the effect, whether the same pattern transfers to other architectures, and which geometric correlations actually matter for learning.

Future Work

The current draft is now substantial enough that the most useful remaining steps are not just "more data," but more targeted validation and cleaner comparisons. Three priorities stand out.

- **Stronger baselines and stronger statistics.** The next paper version should expand beyond matched random sparsity to include pruning-based or learned-mask baselines, and should add fuller confidence intervals or significance analyses across the broader experiment suite.
- **Broader regimes.** The most important empirical question is whether the apparent low-density edge persists on harder datasets, longer training horizons, and additional architectures such as low-density convolutional models, SVHN-style benchmarks, or other non-image tasks.
- **Better geometric analysis.** The current implementation would benefit from richer visual diagnostics and more formal analysis of structured connectivity geometry, including low-discrepancy alternatives, overlap statistics, and dead-unit behavior under different angular and radial rules.

These steps would move the manuscript from a careful exploratory study toward a more comprehensive architectural paper.

A Full Experimental Details

All experiments in this paper were conducted using fixed sparse connectivity masks applied at initialization. The masks were not updated during training. Each experiment was repeated across multiple random seeds to reduce variance and improve reproducibility.

For each dataset, the following protocol was used:

- fixed training epochs for the standard runs, with shorter fixed-step schedules for the edge microbenchmarks;
- matched parameter budgets across dense and sparse models whenever the comparison was intended to isolate connectivity structure;
- identical optimization settings across mask types within each experiment;
- evaluation on both clean and noisy test sets where applicable.

All implementation details, training scripts, and raw logs are available via the linked Colab environment provided at the beginning of the paper.

B Datasets and Architectures

The experiments span multiple datasets and model families to test generality.

MNIST. A simple multilayer perceptron with three fully connected layers was used. This dataset serves as the clearest low-complexity baseline for understanding sparsity behavior.

CIFAR-10. A small convolutional neural network was used to evaluate behavior on a more complex image classification task.

CIFAR-100. A tiny residual-style network was used to test scaling to higher class counts and more challenging distributions.

Edge microbenchmark. A deliberately constrained CIFAR-10 setup with very few training steps and extremely low parameter budgets was used to probe behavior in extreme sparsity regimes.

C Sparse Connectivity Variants

The following connectivity patterns were evaluated:

- **Dense baseline:** standard fully connected or convolutional layers without fixed sparsity;
- **Random sparse:** uniform random selection of active weights under a fixed density;

- **Golden original:** deterministic spiral placement using a golden-angle increment;
- **Aspect-aware golden:** a modified spiral that attempts to normalize for layer aspect ratio.

All sparse variants were matched in total parameter count within each experiment to keep the comparisons fair.

D Additional Results and Per-Seed Tables

All experiments were repeated across multiple seeds. The main paper reports averaged results, while this appendix preserves the full variability and the broader experimental record.

Across datasets, the following patterns were consistently observed:

- the original phyllotactic mask occasionally outperforms random sparsity at very low densities;
- the advantage diminishes or disappears at higher densities or on more complex datasets;
- the aspect-aware variant does not consistently improve performance and often underperforms the original spiral.

Full per-seed results, including accuracy, noise robustness, and intermediate diagnostics, are retained in the supplementary CSV files and notebook outputs linked at the top of the paper.

E Mechanism Diagnostics

To better understand the behavior of different masks, several structural diagnostics were computed:

- **input coverage:** the fraction of input units connected to at least one output;
- **dead outputs:** the fraction of output units with no incoming connections;
- **row and column entropy:** measures of how evenly connections are distributed across the mask;
- **sparsity level:** the proportion of inactive weights.

The results show that simple scalar diagnostics are not sufficient on their own:

- forcing full input coverage does not outperform the original phyllotactic mask;
- random sparse masks can exhibit higher entropy and more uniform distributions than the phyllotactic mask while still performing slightly worse;
- dead outputs matter, but they do not fully explain the observed performance differences.

These observations support a more cautious interpretation: the strongest sparse results in this draft appear to depend on structured connectivity geometry rather than on coverage, entropy, or uniformity in isolation.

F Additional Figures

The appendix can absorb supplementary figures not shown in the main paper:

- mask visualizations for different connectivity patterns;
- additional accuracy curves across epochs or checkpoints;
- extended comparisons across densities and seeds;
- alternative visualizations of connectivity geometry, sparsity, and low-discrepancy structure.

These figures provide further qualitative and quantitative support for the trends discussed in the main text without overloading the main narrative.

G Recovered and Partial Experiments

Some experiments were partially completed or recovered from intermediate checkpoints because of computational and runtime constraints.

These runs are retained for transparency and completeness. While they are not used as the strongest evidence in the main paper, they generally exhibit trends consistent with the fully completed experiments and help document how the final interpretation emerged.

H Implementation Notes

All experiments were implemented using standard deep learning libraries with minimal modifications.

Key design choices include:

- masks applied as fixed multiplicative gates on weight matrices;
- no mask learning or pruning during training;
- consistent initialization and optimization across all variants.

The purpose of these choices was to isolate the effect of connectivity geometry rather than confound it with dynamic sparsification or architecture-specific training tricks.

I Summary of Appendix Findings

The full experimental record reinforces the main conclusions of the paper:

- structured sparsity can help in extreme low-density regimes;
- the effect is not universal and depends on dataset and architecture;
- structured connectivity geometry appears more informative than any single scalar statistic or specific geometric constant used to generate the mask.

This appendix preserves the complete experimental record in prose form while the linked notebook and exported artifacts preserve the detailed code, logs, and per-run outputs.